

Gridfall (Suburban / Urban / Distributed Objectives & Autonomous Forces)

Executive Summary

Gridfall is a subterranean battlefield simulator for distributed objectives, autonomous force coordination, multi-phase control, and distributed environments. The terrain features residential blocks, commercial areas, utility corridors, parks, open areas, cut-de-voes, and a structured power grid that shapes the tactical landscape. The map emphasizes Objective Protocol's autonomous multi-node decision-making, sensor network, routing, control, relay warfare, and tactical access across a dense urban grid. In AR (Real/Latent), the map features a 7-9 m physical distance covering a 30x40 m physical area representing "1 km" of mixed suburban terrain. Key features include distributed objective nodes, multi-phase street networks, interior combat zones, and autonomous force coordination across a 3x4x4 grid.

Design Goals & Gameplay

Distributed Objective Control

Gridfall features distributed control of multiple nodes, subterranean, intersections, relay hubs, and commercial blocks. Victory depends on coordinated multi-phase warfare.

Autonomous Force Coordination

Units operate semi-independently across the grid, selecting optimal nodes to attack or defend based on sensor data, enemy movement, and objective priority.

Urban/Suburban Hybrid Combat

Units navigate streets, corridors, and building layouts, utilizing cover, park, provide open sightlines, strip multi-phase interior flanking routes.

Sensor & Information Warfare

Power outages, sensors, and sensor building layouts disrupt sensors. Relay towers on rooftops restore clarity. Destroying relay creates blind pockets across the grid.

Logistics & Resources

Utility subterranean, commercial depots, and residential garages serve as resupply nodes. Convoys must navigate unpredictable street networks and avoid ambush prone intersections.

Deployment & Progress

A heat map overlay tracks "10,000 points" across players to choose between distributed autonomous squads, heavy urban sensor, or medium drone. Blue displays from the southeast residential block, Red displays from the northeast commercial hub.

Spatial Constraints (Nreal AR)

Visual range: 10-15 degrees FOV, 1000-1500 per eye resolution. Optimal viewing distance 7-9 m.

Playstyle Area & Scaling

10x10 m physical area = 1 km² Suburban grid (3-100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking allows horizontal planes. Depth-mach (if available) provides limited occlusion. Virtual objects render past real-world ones, ensure minimal map-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Residential Block Command) – (0,0) - (100,100) in Center. Cut-de-voes, small parks, main vehicle.

Red HQ (Commercial Hub Command) – (900,900) - (1000,1000) in East. Multi-story, utility poles, industrial structures.

Power Grid Nodes – (100,200) - (900,800) in intersections, along corridors, interior combat zones.

Relay Towers – (100,500) - (900,500) in intersections, along corridors, interior combat zones.

Residential Blocks – (100,100) - (900,100) in intersections, along corridors, interior combat zones.

Parkland Corridor – (500,500) - (500,900) Open landscape, ideal for line-of-sight engagements.

Residential Block East – (500,500) - (900,500) in intersections, along corridors, interior combat zones.

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